

Fluid Dynamics of the Atmosphere and Ocean

Problem Set 2 (Assessed)

This problem set counts 10% toward your final mark. Due on Monday, March 21.
In all cases partial credit will be given if the algebra is wrong but the method right, *provided you explain what you are trying to do.*

1. Vertically propagating Rossby waves

- (a) The quasigeostrophic potential vorticity equation, linearized about a constant eastward mean flow U is

$$\left(\frac{\partial}{\partial t} + U \frac{\partial}{\partial x} \right) q + \beta v = 0,$$

where

$$q = \frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} + \frac{1}{\rho} \frac{\partial}{\partial z} \left(\rho \frac{f_0^2}{N^2} \frac{\partial \psi}{\partial z} \right)$$

is the perturbation potential vorticity. Here, ψ is the perturbation geostrophic stream function, z is a vertical coordinate, f_0 , N and $\beta > 0$ are constants. The density ρ varies in the vertical according to $\rho = \rho_0 \exp(-z/H)$ where ρ_0 and H are constants.

By seeking solutions of the form

$$\psi = \text{Re} \{ \hat{\psi} \exp [i (kx + ly + mz - \omega t) + z/2H] \},$$

where k and l are real, derive the dispersion relation

$$\omega = kU - \frac{k\beta}{k^2 + l^2 + (m^2 + 1/(2H)^2) (f_0^2/N^2)}.$$

What kind of waves are these?

{7}

- (b) Calculate the vertical component of the group velocity for such waves.

{4}

- (c) The perturbation v and θ components will also have wavelike behaviour

$$(v, \theta) = \text{Re} \{ (\hat{v}, \hat{\theta}) \exp [i (kx + ly + mz - \omega t) + z/2H] \},$$

where $g\theta/\theta_0 = f_0 \partial \psi / \partial z$. The northward eddy potential temperature flux is

$$\rho \overline{v\theta} \equiv \rho \left(\frac{1}{L_\lambda} \right) \int_{L_\lambda} v \theta dx$$

where L_λ is one wavelength. Show that

$$\overline{v\theta} = \frac{1}{2} \rho_0 \theta_0 k m \frac{f_0}{g} |\hat{\psi}|^2.$$

Hence explain why upward wave propagation requires a poleward eddy potential temperature flux.

{5}

- (d) Now consider *stationary* waves. Show that upward wave propagation is possible only if U satisfies the inequality

$$\frac{\beta}{k^2 + l^2 + (1/4H^2)(f_0^2/N^2)} > U > 0.$$

{4}

(If the algebra becomes too complicated you may assume $\rho = \text{constant}$ (i.e., $H = \infty$) with only partial loss of credit.)

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2. Eastward jets

1. Show that there can be no extrema of a materially conserved tracer in the interior of a fluid in the presence of a diffusion acting on the tracer (that is, a tracer obeying $D\phi/Dt = \kappa \nabla^2 \phi$ with $\kappa > 0$), if the flow is steady. You may restrict your answer to incompressible flows. {6}
2. Show that the results of part (a) imply that there can be no steady (non-eddy) mean eastward flow at the equator provided that the vertical diffusion of zonal momentum dominates the horizontal diffusion. State and explain carefully any assumptions you make. {6}
3. Show that if there is a source of Rossby waves at any given latitude, we expect that eastward flow will be generated there. {8}

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3. The horizontal velocity in the bottom Ekman layer near the surface at $z = 0$ is governed by

$$-f(v - v_g) = K_M \frac{\partial^2 u}{\partial z^2}, \quad (\text{i})$$

$$f(u - u_g) = K_M \frac{\partial^2 v}{\partial z^2}, \quad (\text{ii})$$

where z is the vertical coordinate and u_g and v_g are the eastward and northward components of the geostrophic velocity.

- (a) Combine equations (i) and (ii) to obtain an equation for the complex quantity $u + iv$. {5}

- (b) Show that the resulting equation has a general solution of the form

$$(u + iv) = (u_g + iv_g) + A \exp \{(1 + i)z/\delta\} + B \exp \{-(1 + i)z/\delta\}$$

where A and B are constants. Give an expression for the depth scale δ . {8}

- (c) Suppose that the boundary conditions are

$$z = 0 : \quad u = v = 0$$

$$z \rightarrow \infty : \quad u = u_g, \quad v = v_g$$

Using the results of the previous parts, or otherwise, obtain a solution for all z . {7}

{20}

4. The shallow water equations are given as

$$\frac{\partial h}{\partial t} + \frac{\partial(uh)}{\partial x} + \frac{\partial(vh)}{\partial y} = 0,$$

$$\frac{D\mathbf{u}}{Dt} + \mathbf{f} \times \mathbf{u} + g\nabla h = 0,$$

where $\mathbf{f} = \mathbf{k}f$ and f is a constant. Obtain an energy conservation law in the form $\partial E/\partial t + \nabla \cdot \mathbf{F} = 0$, and carefully explain why $\mathbf{F} \neq \mathbf{u}E$.

{10}